MARIA LIZETH REYNA-CRUZ, M.D., M.S., Ph.D. Student in Computer Science

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Research Interests

- Explainable Artificial Intelligence (XAI)
- Clinical Decision Support Systems
- Trustworthy Machine Learning in Healthcare
- Medical Imaging and Diagnostic AI
- Uncertainty Quantification in Deep Learning

Education

Present — **University of Texas at El Paso,** El Paso, Texas.
Computer Science PhD. student.

May 2025 — **University of Texas at El Paso,** El Paso, Texas.
M.S. *Computer Science.*

July 2015 — **Universidad Autónoma de Ciudad Juárez,** Ciudad Juárez, Chihuahua.
Medico cirujano.

Publications & Posters

- **Reyna-Cruz, M. L.,** Tabares, R., Ceberio, M., Kreinovich, V., Lauter, C., & Barraza, C. A. M. (2024, October). Machine Learning-Based Screening for Pediatric Hip Dysplasia: Towards a Validated Approach. In *2024 58th Asilomar Conference on Signals, Systems, and Computers* (pp. 1891-1894). IEEE. Available at: https://ieeexplore.ieee.org/abstract/document/10942608?casa_token=Mzo9nAPokTAAAAAA:EC0jr4eGuk-FeIai6VOnzhE1X3LkiyPibLjNwzthhJVqdx-5kPiRRbR_sfq_wxGmDHL-qT2Www

- **Reyna-Cruz, M. L.,** Ceberio, M., Lauter, C., & Lopez Valles, J. M. (2025). Deep Learning Explainability on Non-Hodgkin Lymphoma: Relapse & Treatment. *Proceedings of the Artificial Intelligence in Medicine and Healthcare (AIMH 2025)*, Innsbruck, Austria, p. 76–79. ISBN: 978-84-09-71190-1. Available at: https://sensorsportal.com/AIMH_2025/AIMH_2025_Proceedings.pdf#page=76

- **Reyna-Cruz, M. L.,** Ceberio, M., & Lauter, C. (2025). Trust by Design: Building Reliable Clinical Artificial Intelligence to Advance Quality in Patient-Centered Care. Poster presented at *2025 Forward: Building Trust to Shape the Future of Quality and Human Experience*, Mayo Clinic, Rochester, MN.

- **Reyna-Cruz, M. L.,** Ceberio, M., Lauter, C., Kreinovich, V., and Barraza, C. A. M., ”Efficient First-Approximation Algorithms for Interval-Valued Regression, with Medical Applications in Mind”, *Proceedings of the 2026 Annual Conference of North American Fuzzy Information Processing Society NAFIPS 2026*, El Paso, Texas, March 14–16, 2026, Springer, to appear.

Preprints:

- **Reyna-Cruz, M. L.**, Ruiz-Rohena, K., Guel, Y. I., Salgado, H., Taldir, L., Ramos, E. P., ... & Volkova, A. (2025). Contribution to Error Analysis of Deep Neural Networks: Case of the Activation Functions. Available at: https://inria.hal.science/hal-05367563/

Professional experience

August 2024 – Present

University of Texas at El Paso, El Paso, Texas.
Research Assistant.

- Developed machine learning models for pediatric hip dysplasia screening using ultrasound imaging, achieving clinically relevant diagnostic performance.
- Conducted research on error propagation and robustness in deep neural networks, with focus on activation functions and numerical stability.
- Co-authored peer-reviewed publications in AI for healthcare and presented work at international conferences.
- Collaborated in interdisciplinary teams bridging clinical practice and machine learning deployment.

February 2019 – Present

IMSS (Instituto Mexicano del Seguro Social), Ciudad Juarez, Chihuahua.
Medical doctor.

- Performed medical consultations in clinic and emergency care facility/hospital. Including but not limited to diagnosis, treatment and prevention various physical injuries and diseases.
- Provided emergency care and follow-up treatment in covid patients during the pandemic.
- Performed emergency interventions, including cardiopulmonary resuscitation (CPR) and placement of central venous catheter.
- Assisted at surgeries.
- Developed comprehensive physical exams and patient history to diagnose and control of patients with chronic diseases (diabetes and hypertension).
- Executed prenatal care program and healthy child program.

January – June 2024,
August- December 2025

University of Texas at El Paso, El Paso, Texas.
Teaching Assistant, Object Oriented Programming Class/ Computer Organization/ Operating systems.

- Assisted in preparing course materials, including lectures, quizzes, and assignments.
- Delivered lectures on key programming concepts to undergraduate students.
- Facilitated Demo sessions to reinforce programming concepts for undergraduate students.
- Facilitated review sessions to help students prepare for exams and clarify topics.

- Graded assignments and provided constructive feedback to students.
- Supported students during office hours and lectures, addressing questions and clarifying concepts.
- Participated in the UTEP SEP for Teaching Assistants professional development program, 2025.

December 2017 – October 2020 **Renewed International**, Ciudad Juarez, Chihuahua.
Medical doctor.

- Develop comprehensive physical exams and patient histories. As well as filling of medical files (clinical history, medical notes, medical references) and epidemiological forms (SUIVE).
- Perform initial check-ups and follow-ups of pediatric patients to ensure health throughout the stay at the facility.
- Perform medical consultations. Including but not limited to diagnosis, treatment and prevention various physical injuries and diseases.
- Provide health related lectures to patients.

August 2016 – July 2017 **IMSS Prospera UMR Rancho Colorado**, Rancho Colorado, Bachiniva.
Medical Director of rural clinic.

- Developed comprehensive physical exams and patient history to diagnose and control of patients with chronic diseases (diabetes and hypertension).
- Developed comprehensive physical exams and patient history of patients affiliated to “Prospera” program to ensure health.
- Executed prenatal care program and healthy child program.
- Realized vaccination program.
- Effected first aid care, suture wounds, cleaning wounds, debridement, urinary catheter, orogastric probe, application of intravenous solutions and medicines, and application of intramuscular medicines.
- Performed medical consultations. Including but not limited to diagnosis, treatment and prevention various physical injuries and diseases.
- Provided health related lectures to the community and teenagers.
- Filled medical files (clinical history, medical notes, medical references) and epidemiological forms (SUIVE). As well as other forms (SISPA, OCE99, TC1).
- Developed a work program, a community study, an endemic channels, a collective health diagnosis, and an informational flipchart.

August 2015 – July 2016 **IMSS General Hospital Zone #6**, Ciudad Juarez, Chihuahua.
Intern medic

- Effected first aid care, suture wounds, cleaning wounds, debridement, urinary catheter, orogastric probe, application of intravenous solutions and medicines, and application of intramuscular medicines, take and interpretation of electrocardiogram, splinting.
- Performed CPR.
- Treated and took care of patients at their labour, delivery and immediate postpartum.
- Performed neonatal examination.
- Assisted at surgeries.

- Performed medical consultations. Including but not limited to diagnosis, treatment and prevention various physical injuries and diseases.
- Interpreted laboratory information to provide patients with proper treatment.
- Filled medical files (clinical history, medical notes, medical references) as wells as the census of hospitalized patients.

Awards & Fellowships

- Sloan Centers for Systemic Change Fellow, 2025.
- UTEP Travel Grant Recipient, Artificial Intelligence in Medicine and Healthcare, 2025

Technical Skills

- Programming Languages: Python, Java, R
- Libraries & Frameworks: Jupyter Notebook, TensorFlow, Scikit-learn, Pandas
- Machine Learning & Data Analysis
- Object-Oriented Programming
- Microsoft Office Suite (Word, Excel, PowerPoint)

Languages

- English
- Spanish (Native)